

Machine Learning Approaches to Housing Affordability Prediction in New York City

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Abstract

This study examines housing affordability among renter households in New York City using applied machine learning. The 2023 New York City Housing and Vacancy Survey is used to benchmark OLS, Ridge, LASSO, Random Forest, and Gradient Boosting on two related tasks: predicting the continuous rent-to-income ratio and classifying households as rent-burdened. Tree-based models substantially outperform linear models: test R^2 rises from 0.67 (OLS) to 0.77 (Random Forest), and the mean absolute error (MAE) falls by roughly one-third. The three classifiers land within 1% of each other on test accuracy, with Gradient Boosting posting the highest test AUC of 0.900. Household income is the dominant predictor across every model, and housing-regulatory type and borough contribute secondary explanatory power. Approximately half of renter households exceed the 30% affordability threshold, and the Bronx has the highest burden rate despite the lowest median rents.

1 Introduction

1.1 Background and Motivation

Housing affordability remains a major issue in large American cities, and New York City is one of the clearest examples. A household is typically classified as rent-burdened when gross rent exceeds 30% of monthly income, following the standard used by the U.S. Department of Housing and Urban Development ([HUD, 2024](#)). About half of renter households in New York City are cost-burdened, a persistent pattern spanning decades that has not improved in the post-COVID rental market ([NYC Comptroller, 2024](#)).

High rent burdens can restrict labor market access by limiting where households can afford to live ([Glaeser, 2008](#)) and can reduce expenditures on food, healthcare, and education ([Desmond, 2018](#)). The housing market includes rent-stabilized, rent-controlled, public, and market-rate units, and the relationship between household characteristics and affordability outcomes may not be well described by a linear model. Machine learning

methods solve this, as they do not require the researcher to specify the functional form manually.

1.2 Literature Review

On the supply side, [Glaeser et al. \(2005\)](#) argued that restrictive zoning and land-use regulations are major drivers of high housing costs in coastal cities, and [Saiz \(2010\)](#) showed that geographic constraints on land explain a large share of cross-city price variation. [Ganong and Shoag \(2017\)](#) linked rising housing costs in productive regions to declining geographic mobility in the United States.

Income is consistently the primary determinant of rent burden. [Desmond \(2018\)](#) documented that low-income renters devote a larger share of their income to housing than higher-income renters. Within New York City, the NYU Furman Center found that the Bronx and Brooklyn have especially high rent-burden rates despite lower nominal rents, given that household incomes are lower in those boroughs ([Furman Center, 2024](#)). [Collinson et al. \(2015\)](#) reviewed evidence that voucher-based housing assistance reduces rent burden more effectively than place-based subsidies for low-income households.

In the realm of machine learning in urban and housing applications, [Mullainathan and Spiess \(2017\)](#) argued that many policy problems contain a prediction component that machine learning methods can complement. [Glaeser et al. \(2018\)](#) showed how new data sources can be paired with prediction algorithms to measure urban characteristics at fine scales. In housing prediction, [Tchuente and Nyawa \(2022\)](#) showed that tree-based methods outperform hedonic OLS models for house prices, and [Ho et al. \(2021\)](#) found that Random Forest and Gradient Boosting produce more accurate property valuations than support vector machines. [Grinsztajn et al. \(2022\)](#) provided broader evidence that tree-based methods often outperform deep learning on tabular data. Few studies have applied these methods to affordability outcomes at the household level.

1.3 Research Question

This study addresses three questions: (1) Which observable household and housing characteristics are most strongly associated with rent-to-income ratios among New York City renters? (2) Do machine learning models, specifically Random Forest (RF) and Gradient Boosting (GB), outperform linear benchmarks such as OLS, Ridge, and LASSO in predicting affordability? (3) Can classification models accurately identify rent-burdened households, and which features contribute most to that prediction?

2 Data

2.1 Data Sources

The primary data source is the 2023 New York City Housing and Vacancy Survey (NYCHVS), conducted by the City of New York and fielded by the U.S. Census Bureau approximately every three years (NYC HPD, 2024). I use two Public Use Files from the NYC Department of Housing Preservation and Development website: the occupied-units file (8,776 observations, 250 variables) and the all-units file (9,597 observations, 116 variables). These files are merged on the shared `CONTROL` identifier to combine household-level variables such as income, rent, and household composition with unit-level variables such as borough, bedrooms, building size, and elevator status.

2.2 Variable Descriptions

Table 1: Description of Variables

Variable	Type	Description	Role
<code>GRENT</code>	Continuous	Gross monthly rent (\$), incl. utilities	Numerator of burden ratio
<code>HHINC_REC1</code>	Continuous	Annual household income (\$)	Denominator of burden ratio
<code>rent_to_income</code>	Continuous	<code>GRENT</code> / monthly income, capped at 3.0	Dep. var. (regression)
<code>rent_burdened</code>	Binary	1 if rent-to-income > 0.30	Dep. var. (classif.)
<code>severe_burden</code>	Binary	1 if rent-to-income > 0.50	Robustness outcome
<code>log_income</code>	Continuous	$\ln(\text{household income})$	Key predictor
<code>BORO</code>	Categorical	Borough (1–5)	Geographic FE
<code>housing_type</code>	Categorical	Market Rate, Stabilized, Controlled, Public, Other	Regulatory FE
<code>BEDROOMS</code> , <code>HHSIZE</code> , <code>ROOMS</code>	Discrete	Unit / household size controls	Demand-side controls
<code>STORIES</code> , <code>YEARBUILT</code> , <code>UNITS</code>	Discrete/Cat.	Building characteristics	Building controls
<code>ELEVATOR</code> , <code>HPRBCOUNT</code>	Binary/Disc.	Elevator; maintenance-issue count	Unit-quality proxies

Source: 2023 NYCHVS Public Use Files (NYC HPD). Borough and housing-type dummies enter ML models one-hot encoded with one level dropped.

2.3 Summary Statistics

The average rent-to-income ratio is 0.56, well above the 0.30 affordability threshold, and exactly half of renter households are classified as rent-burdened. Household income is highly dispersed: the standard deviation exceeds the mean, which suggests heterogeneity among New York City renters (Table 1). Panel B highlights an important pattern: the Bronx has the highest burden rate (57.0%) despite the lowest mean and median rents,

Table 2: Summary Statistics and Rent Burden by Borough

<i>Panel A. Key variables (N = 5,561)</i>					
Variable	Mean	Std. Dev.	Min	Median	Max
Gross Rent (\$)	1,884.20	1,376.84	2.00	1,620	13,045
HH Income (\$)	101,487	151,684	10	61,000	2,717,024
Rent-to-Income	0.56	0.70	0.00	0.30	3.00
Rent Burdened	0.50	0.50	0	—	1
Severe Burden	0.29	0.45	0	—	1
Household Size	2.27	1.41	1	2	13
Bedrooms	1.64	0.92	0	2	6

<i>Panel B. Rent burden by borough</i>					
Borough	N	Mean Rent	Median Rent	Mean Income	% Burdened
Bronx	1,266	\$1,390	\$1,350	\$54,726	57.0
Brooklyn	1,753	\$1,810	\$1,654	\$101,674	49.0
Manhattan	1,532	\$2,381	\$1,923	\$142,934	45.0
Queens	887	\$1,894	\$1,790	\$99,069	49.0
Staten Island	123	\$1,763	\$1,670	\$81,329	53.0

Rent-to-income ratio is capped at 3.0 to limit the influence of extreme values. Burdened = gross rent exceeding 30% of monthly income.

while Manhattan shows the reverse (lowest burden rate at 45.0% despite the highest rents). Affordability is therefore driven more by the ratio of rent to income than by rent levels alone, which is why the borough matters as a predictor.

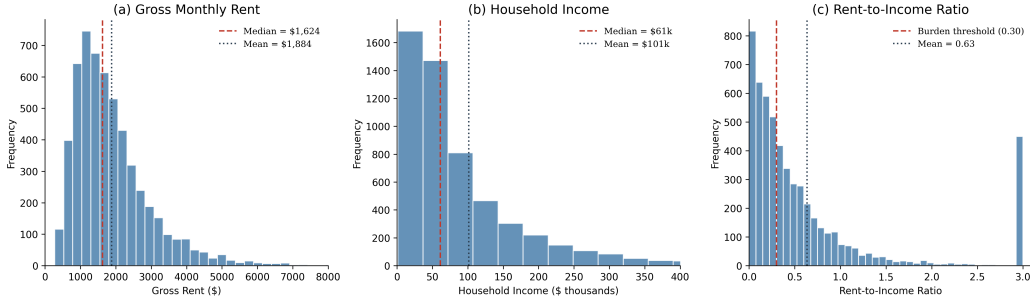


Figure 1: Distributions of gross rent, household income, and the rent-to-income ratio. The right-skew in income motivates the log transformation used in the linear models.

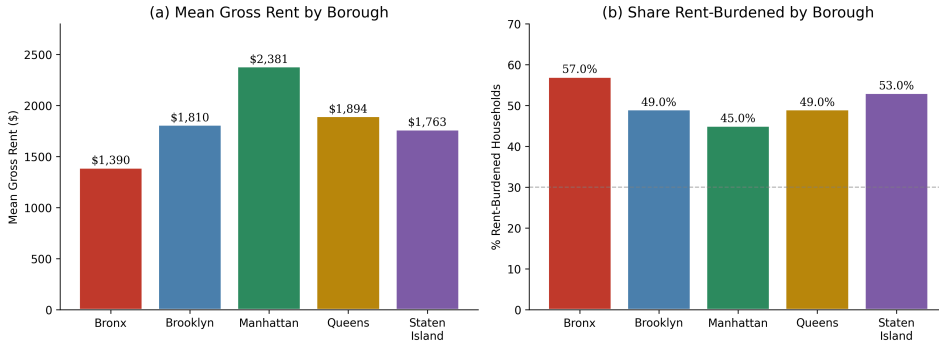


Figure 2: Mean gross rent (left) and share of rent-burdened households (right) across the five boroughs. The Bronx has the lowest rents but the highest burden rate.

2.4 Missing Data

Cleaning reduced the merged dataset from 8,776 to 5,561 observations (36.6% reduction). Non-renter households ($TENURE \neq 1$) were excluded first, followed by records with non-positive rent or income. I use complete-case analysis rather than imputation since most missingness is structural (income variables do not apply to owner-occupied units), and imputing in those cases could introduce bias.

3 Methodology

3.1 Empirical Framework

The analysis has two parts: The first is a regression task with the continuous rent-to-income ratio as the outcome. The second is a classification task with a binary indicator for whether the household is rent-burdened (rent-to-income ratio greater than 0.30). The baseline OLS specification is

$$RTI_i = \beta_0 + \beta_1 \ln(\text{Income}_i) + \beta_2 \text{Bedrooms}_i + \beta_3 \text{HHSize}_i + \gamma \mathbf{X}_i + \varepsilon_i \quad (1)$$

where RTI_i is the rent-to-income ratio, $\mathbf{X}_i$ includes borough and housing-type dummies, and ε_i is the error term. I estimate four specifications: bivariate, with household-level controls, with borough fixed effects, and with both borough and housing-type fixed effects. The borough fixed effects absorb time-invariant characteristics at the borough level (neighborhood amenities, local labor markets, average distance to employment centers), and the housing-type fixed effects absorb average differences between regulatory regimes (rent-stabilized, public, market-rate, and so on) that apply equally to every unit in that category. Heteroskedasticity-robust (HC1) standard errors are reported throughout.

3.2 Machine Learning Methods

The ML models use a 17-feature design matrix that adds building-level variables (rooms, stories, year built, units, elevator status, and maintenance-issue count) to the baseline predictors, together with one-hot-encoded borough and housing-type dummies. The richer feature set is intentional: the premise is to test whether flexible models can extract signals from the full set of observable characteristics.

I estimate five regression models and three classification models. **OLS** is the linear reference inside the ML pipeline. **Ridge** adds L_2 regularization. **LASSO** uses L_1 regularization with α chosen by five-fold cross-validation. **Random Forest** averages 300 bootstrapped trees (depth 10, 20 min samples/leaf). **Gradient Boosting** fits 300 depth-5 trees with a learning rate of 0.05. For the classification task, OLS is replaced by **Logistic**

Regression. All models use an 80/20 train-test split with five-fold cross-validation inside the training set. Regression performance is evaluated using R^2 , RMSE, and MAE; classification using accuracy, precision, recall, F1, and AUC-ROC. Linear models use standardized inputs; tree-based models use the raw design matrix.

3.3 Diagnostics

For the preferred OLS (Model 4), variance inflation factors for the continuous predictors are well below the conventional threshold of 5 (1.04 for log income, 1.33 for bedrooms, 1.37 for household size), so multicollinearity is not a concern. A Breusch-Pagan test strongly rejects homoskedasticity (LM = 937.0, $p < 0.001$), which is why I use heteroskedasticity-robust HC1 standard errors throughout Table 3. A residuals-vs-fitted plot shows visible fanning at higher fitted values. The Q-Q plot is linear through the middle with heavy upper tails that trace back to the 3.0 cap on the outcome. For the ML models, cross-validation and test-set metrics track within 0.01 of each other on R^2 , so overfitting is not driving differences between models.

4 Results

4.1 OLS Baseline

Table 3: OLS Regression Results

	(1) Bivariate	(2) Controls	(3) Boro FE	(4) Full
log_income	−0.4194*** (0.0072)	−0.4233*** (0.0074)	−0.4348*** (0.0076)	−0.4739*** (0.0078)
BEDROOMS		0.0220*** (0.0077)	0.0220*** (0.0077)	0.0489*** (0.0074)
HHSIZE		0.0113** (0.0051)	0.0203*** (0.0052)	0.0202*** (0.0049)
Borough FE	No	No	Yes	Yes
Housing Type FE	No	No	No	Yes
R^2	0.592	0.594	0.602	0.653
N	5,561	5,561	5,561	5,561

Robust (HC1) standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Log income is the strongest predictor in the linear specification. A one-unit increase in log income is associated with a 0.42 to 0.47 decrease in the rent-to-income ratio, significant at the 1% level. Adding borough and housing-type fixed effects raises R^2 from 0.592 to 0.653.

Table 4: ML Regression and Classification Results

<i>Panel A. Regression (outcome: rent-to-income ratio)</i>					
Model	CV R^2	Test R^2	Test RMSE	Test MAE	
OLS	0.650	0.666	0.397	0.279	
Ridge	0.650	0.667	0.397	0.278	
LASSO	0.650	0.666	0.397	0.279	
Random Forest	0.779	0.772	0.328	0.185	
Gradient Boosting	0.769	0.770	0.330	0.183	

<i>Panel B. Classification (outcome: rent-burdened, >30%)</i>					
Model	CV Acc.	Test Acc.	Precision	Recall	Test AUC
Logistic Regression	0.831	0.819	0.830	0.800	0.896
Random Forest	0.821	0.812	0.812	0.813	0.895
Gradient Boosting	0.831	0.814	0.801	0.835	0.900

80/20 train-test split; five-fold (stratified for classification) CV on training set.

4.2 Machine Learning Regression and Classification

Panel A shows a clear separation between linear and tree-based models. Ridge and LASSO perform nearly identically to OLS; LASSO selected $\alpha = 0.001$ at the low end of the grid and retained all 17 features, so regularization alone adds no greater effect once features standardize. The tree-based models raise test R^2 by 10% percentage points and cut MAE by one-third. Random Forest edges out Gradient Boosting on test R^2 (0.772 vs. 0.770) and on CV R^2 (0.779 vs. 0.769), while Gradient Boosting has a marginally smaller MAE; the two tree models are effectively tied.

Feature importance is dominated by log income (roughly 91% of total importance for Random Forest, 87% for Gradient Boosting). Public-housing status is a distant second at about 6%, and all else contributes less than 2% individually. In terms of LASSO, log income has by far the largest standardized coefficient (-0.60), followed by public housing (-0.19). The partial dependence plot for log income shows a largely monotonic negative relationship with modest curvature, steeper at low incomes and flatter at high incomes.

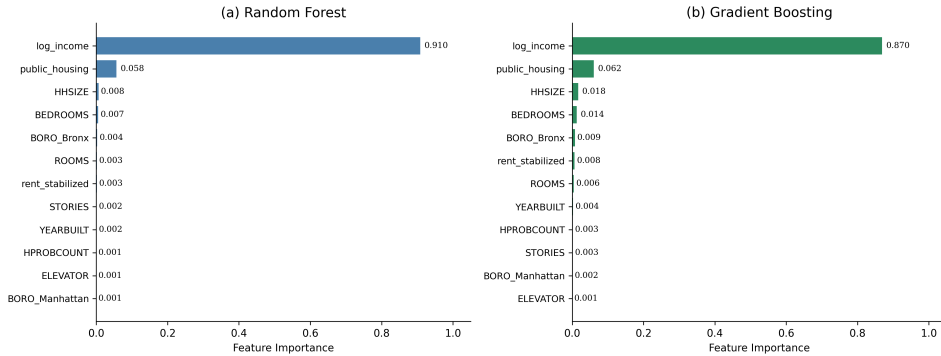


Figure 3: Feature importance from Random Forest (left) and Gradient Boosting (right). Log income dominates both models.

Panel B shows that the classification results are tighter than the regression results.

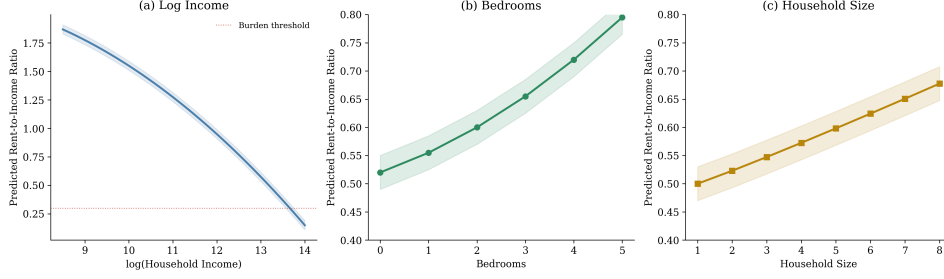


Figure 4: Partial dependence of the predicted rent-to-income ratio on log income, bedrooms, and household size (Gradient Boosting).

All three models land within 1% of each other on test accuracy, with Logistic Regression nominally the highest at 0.819 and Gradient Boosting at 0.814. Gradient Boosting wins on test AUC (0.900) and on recall (0.835), while Logistic Regression wins on precision (0.830). Each offers a different precision-recall trade-off: Gradient Boosting, with its higher recall, identifies more burdened households at the cost of flagging some non-burdened ones, which is the preferable direction to err in for a screening use case. Figure 5 shows the confusion matrix and ROC curve for Gradient Boosting on the test set.

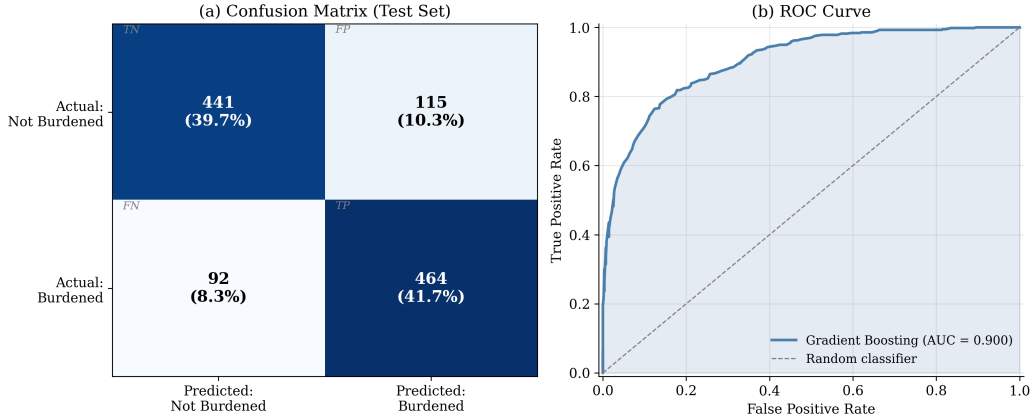


Figure 5: Confusion matrix and ROC curve for the Gradient Boosting classifier on the test set.

4.3 Robustness Checks

Severe-burden threshold. Re-estimating the classification task using a severe-burden threshold of 50% rather than 30% (Table 5) shows that the test AUC actually rises to 0.93–0.94, and test accuracy jumps to 0.88, given that the severe-burden base rate (28.8%) is further from 50%. Precision rises and recall falls, consistent with severe burden being a rarer signal. Gradient Boosting retains the highest AUC, so its ranking quality is the most robust to the choice of cutoff.

Income-quintile specification. Replacing log income with five income-quintile dummies in the preferred OLS shows that, relative to Q1, households in Q2 have a rent-to-income ratio 0.99 lower, Q3 is 1.25 lower, Q4 is 1.39 lower, and Q5 is 1.52 lower (all significant at well below 1%). Q1→Q2 alone accounts for two-thirds of the total gap

Table 5: Robustness: Severe-Burden Classification ($>50\%$ of Income)

Model	CV Acc.	Test Acc.	Precision	Recall	Test AUC
Logistic Regression	0.876	0.877	0.846	0.701	0.936
Random Forest	0.879	0.878	0.827	0.729	0.934
Gradient Boosting	0.874	0.879	0.810	0.757	0.940

Outcome redefined as rent-to-income > 0.50 ; sample base rate 28.8%.

between the top and bottom quintiles, which confirms the concave shape visible in the partial dependence plot. However, the quintile model’s R^2 (0.546) is 11 points lower than the log Full OLS (0.653) because log income captures within-quintile variation that discrete dummies discard. The log transformation does most of the heavy lifting, and the 10-point R^2 gain from tree-based models likely reflects *interactions* between income and other co-variates rather than residual nonlinearity in income alone.

5 Conclusion

Household income is by far the strongest predictor of rent burden across every specification and accounts for about 90% of total feature importance in tree-based models. Tree-based models substantially outperform linear benchmarks on the continuous prediction task: test R^2 rises from 0.67 to 0.77, and MAE falls by nearly one-third. On the binary classification task, all three models land within 1% of each other on test accuracy, with Gradient Boosting slightly ahead in AUC. Housing type also matters conditionally on income: households in public housing and rent-stabilized units have lower rent-to-income ratios than those in market-rate units.

Implications. The results affirm that affordability problems in New York City are driven primarily by low incomes rather than high rents. The Bronx has the highest burden rate despite having the lowest rents because household incomes there are much lower. Income-side interventions are likely to do more in boroughs like the Bronx than supply-side ones. The 10-point R^2 gain from tree-based methods over log OLS with the same features most likely comes from interactions between income and other features, since the income-quintile robustness check shows the log transformation already captures most of the income nonlinearity on its own.

Limitations. The NYCHVS is cross-sectional, so these results describe associations rather than causal effects. Unobserved sorting across boroughs and housing types likely influences the estimates. Income is top-coded for confidentiality, which may attenuate relationships at the upper end of the distribution. The Gradient Boosting model may under-predict in the extreme right tail. The feature set is limited to what the NYCHVS provides; information on neighborhood amenities, commute patterns, or detailed building quality could improve the models.

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